FLUTTER LABORATORY ACTIVITY 1

Creating a Student Profile Using StatelessWidget

Estimated Time: 1–2 hours

Activity Type: Individual

Output: A Flutter application displaying a student profile

I. Activity Overview

In this activity, you will create a simple **Student Profile Application** using Flutter. Your application will display your name, student number, course, year level, email address, hobbies, and a short introduction.

You will use StatelessWidget to build the interface. A StatelessWidget does not manage mutable internal state, making it suitable for this profile screen with predefined information. Dart API

II. Learning Objectives

At the end of this activity, you should be able to:

1. Create and run a Flutter project.
2. Explain the purpose of StatelessWidget.
3. Arrange widgets using Column, Padding, and SizedBox.
4. Display information using Text, Icon, CircleAvatar, and Card.
5. Customize the appearance of a student profile.

Upload it here the output for the student profile activity.

Application Requirements

Your application must display:

1. An application bar titled **Student Profile**.
2. A circular avatar containing your initials.
3. Your full name.
4. Your course or program.
5. Your student number.
6. Your year level and section.
7. Your school email address or a sample email address.
8. A short **About Me** paragraph.
9. At least three hobbies or interests.

Important: All classes you create for the interface must extend StatelessWidget. Do not use StatefulWidget or setState() in this activity.

Step 1: Create a Flutter Project

In the terminal, enter:

```
flutter create student_profile_app
```

Move into the project folder:

```
cd student_profile_app
```

Open the folder in Visual Studio Code:

```
code .
```

If the code command is unavailable, open Visual Studio Code manually and select **File** → **Open Folder**, then choose student_profile_app.
Checkpoint: Your editor displays the Flutter project folders.

Step 2: Locate the Main Dart File

Inside your project:

1. Open the lib folder.
2. Open main.dart.
3. Remove the starter code.
4. Save the file.

You will replace the sample counter application with your student profile.

Step 3 : Understand the Application Structure

Your application will contain two custom widgets:

Widget	Purpose
StudentProfileApp	Configures the application title, theme, and home screen.
StudentProfilePage	Builds the student profile interface.

Both classes will extend StatelessWidget.
The main parts of the screen are:

Flutter Widget	Purpose in This Activity
MaterialApp	Provides the Material application structure.
Scaffold	Provides the screen layout.
AppBar	Displays the screen title.
SingleChildScrollView	Allows scrolling when the content exceeds the screen height.
Column	Arranges widgets vertically.
CircleAvatar	Displays the student's initials inside a circle.
Card	Groups related profile details.
ListTile	Displays an icon, label, and detail.
SizedBox	Adds spacing between widgets.

Step 5: Write the Application Code

Copy the following code into lib/main.dart:

```
import 'package:flutter/material.dart';
import 'student_profile_app.dart';

void main() {
  runApp(const StudentProfileApp());
}
```

Step 5.1: Create lib/student_profile_app.dart

This file contains the application configuration and theme.

```
import 'package:flutter/material.dart';
import 'student_profile_page.dart';

class StudentProfileApp extends StatelessWidget {
  const StudentProfileApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Profile',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: Colors.indigo,
        ),
        useMaterial3: true,
      ),
      home: const StudentProfilePage(),
    );
  }
}
```

Step 5.2: Create lib/student_profile_page.dart

This file contains the student profile screen.

```
import 'package:flutter/material.dart';

class StudentProfilePage extends StatelessWidget {
  const StudentProfilePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color(0xFFF3F4F8),
      appBar: AppBar(
        title: const Text('Student Profile'),
        centerTitle: true,
        backgroundColor: Colors.indigo,
        foregroundColor: Colors.white,
      ),
      body: SafeArea(
        child: SingleChildScrollView(
          child: Padding(
            padding: const EdgeInsets.all(24),
            child: Column(
```

```
    ),
    body: SafeArea(
      child: SingleChildScrollView(
        child: Padding(
          padding: const EdgeInsets.all(24),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: [
              // Profile avatar.
              const Center(
                child: CircleAvatar(
                  radius: 55,
                  backgroundColor: Colors.indigo,
                  child: Text(
                    'JD',
                    style: TextStyle(
                      fontSize: 36,
                      fontWeight: FontWeight.bold,
                      color: Colors.white,
                    ),
                  ),
                ),
              ),

              const SizedBox(height: 16),
```

```
              const SizedBox(height: 16),

              // Student name and course.
              const Text(
                'Juan Dela Cruz',
                textAlign: TextAlign.center,
                style: TextStyle(
                  fontSize: 26,
                  fontWeight: FontWeight.bold,
                  color: Colors.indigo,
                ),
              ),

              const SizedBox(height: 8),

              const Text(
                'BS Information Technology',
                textAlign: TextAlign.center,
                style: TextStyle(
                  fontSize: 16,
                  color: Colors.black54,
                ),
              ),

              const SizedBox(height: 24),
```

```
const SizedBox(height: 24),

// Student details.
const Card(
  child: Column(
    children: [
      ListTile(
        leading: Icon(
          Icons.badge,
          color: Colors.indigo,
        ),
        title: Text('Student Number'),
        subtitle: Text('2026-00123'),
      ),
      Divider(height: 1),
      ListTile(
        leading: Icon(
          Icons.school,
          color: Colors.indigo,
        ),
        title: Text('Year Level and Section'),
        subtitle: Text('2nd Year - IT2A'),
      ),
      Divider(height: 1),
      ListTile(
        leading: Icon(
```

```
      Divider(height: 1),
      ListTile(
        leading: Icon(
          Icons.school,
          color: Colors.indigo,
        ),
        title: Text('Year Level and Section'),
        subtitle: Text('2nd Year - IT2A'),
      ),
      Divider(height: 1),
      ListTile(
        leading: Icon(
          Icons.email,
          color: Colors.indigo,
        ),
        title: Text('Email Address'),
        subtitle: Text('juan@example.com'),
      ),
    ],
  ),
),

const SizedBox(height: 24),
```

```
            // Introduction.
            const Text(
              'About Me',
              style: TextStyle(
                fontSize: 20,
                fontWeight: FontWeight.bold,
              ),
            ),

            const SizedBox(height: 8),

            const Text(
              'I am an IT student who enjoys learning '
              'about technology and creating applications. '
              'My goal is to become a mobile developer.',
              style: TextStyle(
                fontSize: 16,
                height: 1.5,
              ),
            ),

            const SizedBox(height: 24),

            // Hobbies.
```

```
            // Hobbies.
            const Text(
              'Hobbies and Interests',
              style: TextStyle(
                fontSize: 20,
                fontWeight: FontWeight.bold,
              ),
            ),

            const SizedBox(height: 8),

            const Wrap(
              spacing: 8,
              runSpacing: 8,
              children: [
                Chip(label: Text('Coding')),
                Chip(label: Text('Reading')),
                Chip(label: Text('Badminton')),
              ],
            ),
          ],
        ),
      ),
    ),
  ),
);
```

```
                ),
              ),

              const SizedBox(height: 8),

              const Wrap(
                spacing: 8,
                runSpacing: 8,
                children: [
                  Chip(label: Text('Coding')),
                  Chip(label: Text('Reading')),
                  Chip(label: Text('Badminton')),
                ],
              ),
            ],
          ),
        ),
      ),
    ),
  );
 }
}
```

Step 6: Run the Application

Save main.dart, then check the available devices:

```
flutter devices
```

Start your emulator or connect your device, then run:

```
flutter run
```

If you are using Chrome and it appears in the available devices, enter:

```
flutter run -d chrome
```

Expected Result: The application displays a student profile with an indigo app bar, circular initials, student details, an introduction, and hobby labels.

Step 7: Personalize Your Profile

Update the sample information with your own details.

Sample Content	Replace With
JD	Your initials
Juan Dela Cruz	Your full name
BS Information Technology	Your course or program
2026-00123	Your student number
2nd Year - IT2A	Your year level and section
juan@example.com	Your school email or a sample email
Sample introduction	Your own introduction of 2–3 sentences
Coding, Reading, Badminton	At least three of your hobbies

Save the file and use **Hot Reload** to view your changes.
Reminder: Editing the source code and using Hot Reload does not require setState().

VI. Guide Questions

Answer each question in 2–3 sentences.

1. What is a StatelessWidget, and why is it appropriate for this application?
2. What does the build() method return?
3. What is the difference between MaterialApp and Scaffold?
4. How do Column, Padding, and SizedBox help organize your screen?
5. Why did you use SingleChildScrollView in the profile page?

VII. Submission Instructions

Upload the following to **BBL**:

1. **GitHub link**

```
git link
```

1. Keep the original file named main.dart inside your Flutter project.
2. **Screenshot of the running application**
3. Make sure your name and profile details are visible. Submit additional screenshots if the screen requires scrolling.

```
LastName_FirstName_Lab1_Output.png
```

1. **Answers to the guide questions**
2. Enter your numbered answers in the BBL submission text box. If text entry is unavailable, attach a .txt file.

Before clicking Submit: Open your attachments and check that you uploaded the correct files.

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VIII. Grading Rubric

Criteria	Points
Application runs successfully	20
Correct use of StatelessWidget	20
Complete student profile information	25
Readable layout and at least three visual changes	15
Answers to guide questions	10
Complete submission files	10
Total	**100**